Cobble Logging User's Guide

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This document details the use of a function for recording and analyzing mill parameters during a period of time immediately preceeding a trigger (usually a mill cobble). Graphical and statistical analysis tools are included in this function.

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1 Description

The GE Cobble Logging system (COBLOG) provides the capability to record and analyze mill parameters during a period of time immediately preceeding some trigger (usually a mill cobble). The data can then be displayed and analyzed utilizing interactive graphical analysis, which includes statistical information such as correlation, regression, standard deviations, and cubic spline curve fitting.

Parameters to be recorded can be local DMC variables, which are sampled by each individual DMC, or CSF global variables, which can be sampled by the Supervisory VAX or any DMC. In either case, the parameters are sampled continuously at some fixed interval and saved in a circular list within the local memory of each processor.

When a trigger occurs, sampling continues for one second, and then all the lists are frozen. The circular lists from all of the DMCs are then automatically uploaded via the CSF and stored, along with the VAX circular list, on the Supervisory VAX disk. The fault trigger is derived by logic in one or more of the DMC controllers. A facility is also provided for the user to generate a trigger on demand to record the current circular list contents. Upload time is approximately 10 seconds per DMC.

In each DMC, the sample rate can be up to the fastest segment scan rate in the DMC, if desired (typical sample rates are 10-20 msec). Different DMCs may have different sample rates, depending on their function. Up to 8 real and 14 logical parameters may be recorded in each DMC, and these values do NOT need to be CSF global signals; they can be any variable within the local DMC. The circular list is 16K bytes; the number of variables collected and the sample rate will determine the span (in time) of the data.

In the Supervisory VAX, up to 50 variables (any combination of real and logical) may be collected at a fixed sample rate of 100 msec. Exactly 200 samples are maintained for each variable, giving 20 seconds of data for each trigger. Parameters which are collected by the VAX must be CSF global signals.

Analysis of the data is accomplished using a highly-interactive graphical package which allows the user to predefine the variables of interest which are to be displayed together on each screen. Up to four graphs may be displayed on a single screen, with up to 4 real (or 8 logical) values per graph. Each screen is described by a "plot definition file" which defines the variable layout, default scaling information, textual titles, etc. Once a given screen had been displayed, the data on that screen can be rescaled, correlated, or displayed numerically if desired.

A database of uploaded cobble data is maintained, indexed by the date and time of the trigger. Data from previous events can be recalled and analyzed as desired, using the current set of plot definition files. Plot definition files can be "chained" to allow the user to page through a set of analysis displays looking for a specific problem or event. Multiple terminals may be analyzing the same cobble data at the same time.

2 Requirements

Graphics Terminal

The graphical analysis tool requires a color graphics terminal. Currently supported devices are DEC's VT340, and the IDT 2250.

Control Signal Freeway

The COBLOG software uploads circular data lists from each DMC by issuing "directed messages" on the Control Signal Freeway (CSF). The associated CSF hardware and software must be present for COBLOG to function.

Data Storage

Data storage requirements for each cobble are small; while there is no limit on stored cobbles, only 64 can be displayed by the plotting function, therefore the user must assure 64 or less in a directory. The user may purge them as desired. The 14 logical and 8 real limits per DMC are the recommended limits, but these are not enforced by the CBL block (e.g.: 25 reals may be recorded, but the span will only be a few seconds at a fast scan rate...).

DMC Configuration

Segment #1 is used in the DMC for the 16K circular list, which is always present. However, if a particular DMC is using the logic accelerator (which uses segment 1), an additional 16K of memory must be added to the DMC to support cobble logging.

3 Operator Interface

The Cobble Logging system has three major parts. An online function, resident in each control product, is responsible for collecting its own local data and maintaining an internal circular list. A second VAX function can, upon request, upload these circular lists from one or more control products into VAX disk files for analysis and processing. The third function then converts the uploaded raw data into graphical plots and displays them. It is this third function which provides the primary operator interface to the "cobble" data. The following sections detail how the operator can specify a plot, and then how to use the provided PLOT utility to graphically analyze the data.

3.1 PLOT Specification Language

This chapter details how to create a plot specification file, which is used to specify plotting screens for the PLOT utility.

The user can define which variables will be displayed together on each screen by creating input data files for the PLOT analysis function. Each screen may contain a practical limit of four plots, and each plot may contain up to four variables each (or up to 8 logicals), for a total of 16 real data variables (or 32 logical variables) on a single screen.

The user can interactively define the starting and ending date and time for the plot, within the range of data points which are present on the disk.

3.1.1 Sample Plot Descriptor File

The following is a sample of a plot descriptor file. The file is free-format; each line may contain any number of tabs and/or spaces between keywords and parameters and comments. Blank lines are permitted for readability, and any text following a "!" character is ignored (for comments). Some of the commands are optional, as noted in the file comments.

All lower_case text in the example denotes parameters which the user must define.

3.1.1.1 Sample File TEST.PLOT

```
! ---------------------------------------------------------------------
TITLE "plot_title"              ! optional (defaults to "plot_name")
X_SCALE min max                 ! optional (program-supplied defaults)

PLOT_1 min max units            ! required (units optional)
  DATA name                     ! required
  DATA name "description"       ! required (description optional)
     !      .
     !      .   (up to four per real plot)
     !      .   (if min & max are both 0.0, then program will autoscale)

PLOT_3 Logical                  ! required
  DATA name                     ! required
  DATA name                     ! required
     !      .
     !      .   (up to eight per logical plot)

DISPLAY PLOT                    ! required to actually draw the plots

INCLUDE plot_name               ! to chain to another file
! ---------------------------------------------------------------------
```

3.1.2 Plot File Commands

The following commands are supported in a plot definition file:

- TITLE "text"

  This command defines a title for the graph. If you do not specify a title, the program uses the name of the plot definition file as the title. If your title contains lower case text, you must enclose it in quotes or it will be converted to upper case.

- X_SCALE min max

  This command defines the scaling for the x-axis. "min" and "max" are real numbers which define the range of times you wish to sample. Units are always seconds.

- PLOT_n min max "units"

  This command starts a real subplot (one box on the screen). Subplots must be defined in order (n=1-4). The parameters define the y-axis scaling for the subplot (real values). Specifying "min" and "max" as both 0.0 will cause the plotting routine to select appropriate values so that all the data is displayed (autoscaling). Up to four real data points can be specified in each subplot. The "units" qualifier specifies text for the y-axis of the plot.

- PLOT_n LOGICAL

  This command starts a logical subplot. Subplots must be defined in order (n=1-4), with any combination of real and logical subplots together on a screen. There are no scaling values for a logical subplot; only the keyword "LOGICAL", which may be abbreviated to a single "L" if desired. Up to eight logical data points can be specified in each logical subplot.

- DATA variable_name "description"

  The DATA statement defines a variable to be plotted. This selects a variable as well as specifying an optional description with which to tag each plot curve (the "description" field). The convention for specifying a variable name to the COBLOG utility is to place the DMC name followed by a period followed by the local variable name (e.g. XYZ.local_name). If the variable is found on the VAX, use "VAX" as the prefix name (e.g. VAX.local_name). If the "description" field is omitted, the variable name itself is printed to the right of each curve.

- DISPLAY PLOT

  This command causes the plot to be drawn on the screen. This must be included in at least one plot file to create a plot. It allows several screens to be specified in one plot file, or several plot files to be chained together.

- "INCLUDE plot_file"

  This command causes the parser to begin reading from the defined plot file. The ".PLOT" file extension is assumed if no extension is present. This command allows you to "nest" plot definitions as desired. For example, if you have several single plots defined, you could define a file which contained a list of "INCLUDE" statements which would then cause the plots to be drawn one after the other. The user must select the "P" option in each plot to advance to the next plot in the list.

The user can also control the defaults which are applied to any parameters which are not specified in a plot definition file. Define a file called "_DEFAULTS.PLOT" by following the same format as a plot file. This file is read by the parser to initialize each plot.

3.2 Graphical Analysis

The Cobble Logging Plotting function reads the uploaded data and generates user-defined plots. The user may select up to four real or count variables (up to 8 logicals) to be plotted together on a single subplot, and up to four subplots per screen (16 real variables, 32 logicals, or a mixture of both per each graphics screen). This function must be initiated by the user; it requires user input to define which set of cobble data (by date) is to be plotted, and which variables will be plotted together on which graphs. The output from the plotting function will be X-Y plots on a VT340 or IDT terminal.

The plotting function is an interactive utility which uses menus and prompts to lead the user through the process of generating a set of plots of the various data values which have been collected. The plots will be labeled with the names of each variable plotted and the date and time that the fault trigger was seen by the upload function.

The user is presented with a menu of all the cobble times for which data is saved on disk (in reverse chronological order). After selecting a particular cobble date and time, a menu of predefined plots (limit of 64) is presented. Plots can be defined by creating a plot definition file (using any DEC editor), which defines the variables to be plotted together on a single screen, some scaling information, and other plot parameters. The name of the plot definition file will be the "name" of that plot screen; the user may create up to 64 of these plot files. The capability to "chain" several plot files together to form a logical sequence of screens is also provided.

Since this function is an interactive utility, all errors are displayed to the user at runtime. If a predefined plot file requests plotting of some data point which was not collected for the current cobble, the corresponding plot on the display will be blank and contain an appropriate error message.

The following sections describe the actual PLOT utility with which the user interactively analyses their plotting data. The purpose and function of each interactive command is presented.

3.2.1 Purpose

The PLOT utility is designed to provide a data analysis and plotting function in one easy to use utility. User interaction is crucial to allowing flexible data analysis. A statistics mode of operation is provided wherein the user is presented with a list of all non-logical variables from the main plotting screen. The user selects two variables to generate cross-correlation statistics, or the same variable is selected twice to perform auto-correlation statistics. Two subplots are used to graphically present the two selected variables to the user. A dotted red line is displayed on the same subplot with each curve to depict the regression line resulting from the scaled data.

For nonlinear datasets, the regression line (and its coefficients) change as the user changes the plot x-axis scaling. This allows for choosing the exact time interval in which to analyze the data. On either the main screen, the expanded mode, or the statistics mode, the user may select a function where the data points (x and y values) are tabulated to the screen. This helps to prevent misinterpretation of the data.

3.2.1.1 Available Commands

The available commands are presented according to what "mode" the PLOT utility is in. Two features have been added to aid in the entering of commands. Many prompts display a default value along side the prompt. If the user wishes to accept the default value, merely press the RETURN key. If a new value is to be entered, enter the value before pressing RETURN and the default value is overridden. The second feature added to the interactive plot analysis function is the ability to perform "type-ahead". If a long, slow plot is being displayed, for example, and the user wishes to interrupt this action with a new command, simply press the indicated key for that command (e.g. "D" will enter display numbers mode) and after a brief delay the existing command will be replaced by the new one without waiting for the original command to complete. Users will appreciate the time-saving aspects of this feature. Commands associated with the main screen are described first, followed by commands accessible from the expanded mode, the display numbers mode, and finally the statistics mode.

3.2.1.1.1 MAIN SCREEN

The main screen is the screen the user first sees when entering the PLOT utility invoking a particular plotting file (see the chapter on the PLOT specification language). This screen limits the number of variables the user may simultaneously analyze. The following commands are accesible from this main screen.

3.2.1.1.1.1 Help Option

This option clears the current plotting screen and displays a list of the commands available from the main screen, along with a short description of each. After reviewing the text, the user may press any key and the screen is returned and redrawn in the mode from which the HELP option was requested.

3.2.1.1.1.2 Change Scaling

This option allows changing the X or Y scaling, or resetting the plot to the original scale defined in the plot specification file. On the main screen, all subplots have the same x-axis scaling, to allow comparing curves at identical time intervals. Therefore, if the user changes the x-axis by specifying a new XMIN and XMAX (units of time), all subplots are redisplayed when the user presses the UPDATE key. If, however, the user wishes to change y-axis scaling, the PLOT utility prompts the user for which subplot to change, since subplots with differing y-axis units may be

displayed on the same plotting screen. Scaling changes do not immediately refresh the screen. See the following description of the "Update Display" function.

3.2.1.1.1.3 Update Display

Scaling changes do not immediately result in an updated plot. This is done to allow users to make multiple scaling changes before waiting for the sometimes tedious graphics screen refresh process to complete. After the user has made the desired scale changes, they simply press the "U" key to update the display using the scaling as changed.

3.2.1.1.1.4 Expanded Mode

Pressing the "X" key is used to enter the expanded mode. In expanded mode, the user is prompted for one subplot (subplots are numbered 1-4 from the top of the screen). After entering their choice, the screen is cleared and the selected subplot is drawn to fill the screen. This option is useful only if multiple subplots are on the original plotting screen, and the user needs more graphics resolution to analyze a plot. The expanded mode plot is drawn with more grid lines (and labeled x and y-axis tick marks) than in the normal, smaller subplots.

To exit from the expanded mode, the user presses "E" for Exit.

3.2.1.1.1.5 Display Numbers Mode

Entering a "D" from the main menu places the user in display numbers mode, where (x,y) data values are displayed in a tabular format. See section Section 3.2.1.1.3 for a description of this mode.

3.2.1.1.1.6 Statistics Mode

Entering an "S" from the main menu places the user in the statistics mode. This mode, and the associated commands, are displayed in section Section 3.2.1.1.4.

3.2.1.1.1.7 Next Plot File

Entering a "P" exits the user from the control of the current plotting "screen" and, if multiple screens are defined for the current plot specification file, proceeds immediately to the next specified plotting screen. If no further screen is specified, this key returns the user to the prompt for a new plot file name.

3.2.1.1.1.8 Abort to Menu

Pressing the "M" key immediately aborts the PLOT utility session, leaving the cobble logging utility.

3.2.1.1.2 EXPANDED MODE

Pressing the "X" key from the main screen places the user in the expanded mode, as described above. The available commands in this mode are the same as in the main menu, with the following exception.

3.2.1.1.2.1 Exit

"E" exits the expanded subplot mode and returns to the main screen.

3.2.1.1.3 DISPLAY NUMBERS MODE

At any time, in the main screen or expanded mode, the user may enter a mode in which the x and y data values are drawn to the screen in tabular format. The display numbers mode first prompts the user for which subplot to display numbers. After the user responds, the graphics screen is cleared and a small representation of the chosen subplot is drawn to the top of the new graphics screen. Then, using color coding based on the color with which each variable's curve is drawn, the x & y data pairs are displayed on the screen. For each variable, the screen displays 100 (x,y) data pairs at a time and pauses until the user presses any key. For subplots with multiple variables, the next variable is displayed in its color after the numbers from the first variable have been displayed.

The user has control over how the numbers are displayed. To immediately scroll to the next "block" of 100 numbers, the user may press the enter key before the data has been drawn. If the user wishes to immediately proceed to the next variable on a subplot, the "N" key may be pressed. If the user is ready to exit the display numbers mode, the "E" key may be pressed.

3.2.1.1.4 STATISTICS MODE

The statistics mode has been designed separately from the main screen mode to try to provide similar functionality, while at the same time maximizing the ease with which a user can perform auto and cross-correlations on data.

The two curves being analyzed are displayed in two subplots at the top of the plotting screen. A linear regression line is fitted to the data of each curve, and displayed as a dotted red line "underneath" the curve. The coefficients for this line (slope and y-intercept) are displayed to the right of each subplot. Should the number of points plotted fall below 50 points, a small box will be drawn at the location of the SAMPLED data points for that curve, and a cubic spline curve fitting algorithm is then used to generate a solid red curve which passes through each of the SAMPLED data points. The term "SAMPLED" refers to the algorithm used to obtain a one-to-one correspondence between data points from two curves. If one

curve samples data every 30 milliseconds for example, and the second curve samples data every 50 milliseconds, then the data points will not be coincident for a given x-axis time. The "SAMPLING" algorithm finds which curve has the least number of points, and then chooses the same number of points from the second curve, choosing the points closest to the desired "base" curve. This sampling can create "aliasing" problems in rapidly varying data, therefore the user should refer the boxes drawn when the analysis is performed on less than 50 data points. In addition to the linear regression coefficients, the arithmetic mean (average) and the standard deviation are displayed alongside each curve.

3.2.1.1.4.1 Help Option

Pressing the "H" key invokes the help option. This option performs like its counterpart in the main screen mode, but details the commands available from the statistics mode.

3.2.1.1.4.2 Change Scaling

Pressing the "C" key invokes the Change scaling option. The x or y axis scaling of either statistics subplot (#1 or #2) may be changed, or the axes may be reset to the default scaling values. When changing scaling, the user is prompted for the axis and subplot information. The only significant departure between changing scaling in the statistics mode and changing scaling in the main PLOT mode is how the default scale values are presented. In the main plotting mode, the user is always presented a default value equal to the current axis limits the user is changing. In the statistics mode, however, the process is somewhat different. The main goal of changing scaling in the statistics mode is to allow the user to achieve identical x-axis durations on both subplots in order to obtain an auto- or cross-correlation. At the same time, however, the user might also want to examine all the data from one curve and compare it to a small section from another curve. To provide the flexibility needed, the user is always presented default XMAX values on the x-axis that will result in identical durations between the subplot the user is changing and the other subplot. If the default XMAX would result in exceeding the available data, however, the user is presented with a default equal to the end of the available data, and a warning is printed.

The y-axis scaling does not influence the statistics in any way. Changing y-axis scaling is provided only as a convenience to the user in order to help visualize the data being analyzed. It should be pointed out that correlation analysis is not affected by the scaling of the y-axis because any scale factor is incorporated in both the numerator and denominator of the correlation equation, thus cancelling its effect.

3.2.1.1.4.3 Update Display

As in the main display mode, the "U" key is used to signal the end of any scaling changes, at which time the screen may now update the curves, and if data is available for two identical x-axis durations, display the regression line for each curve and the correlation factor.

3.2.1.1.4.4 Display Numbers

The display numbers option of the statistics mode is different from its counterpart in the main plot mode in that it does not erase the entire screen to display a tabulation of 100 numbers. Instead, this option has been modified to leave the two regression subplots on the screen, prompt the user for which subplot to display numbers, and then display 40 (x,y) pairs using the lower half of the plotting screen. As with the normal mode display numbers option, the user may press the enter key to update to the next block of 40 numbers before the prior set has finished displaying. Pressing the "E" key after any given set of 40 values has been displayed allows the user to Exit the display numbers mode.

3.2.1.1.4.5 Exit

The "E" key is used to exit the statistics mode and return the user to the mode from which statistics was entered.

3.2.1.1.4.6 Next Plot File

The "P" key aborts the interaction with the current screen, and if another plotting screen has been defined, starts data collection for this next screen.

3.2.1.1.4.7 Abort to Menu

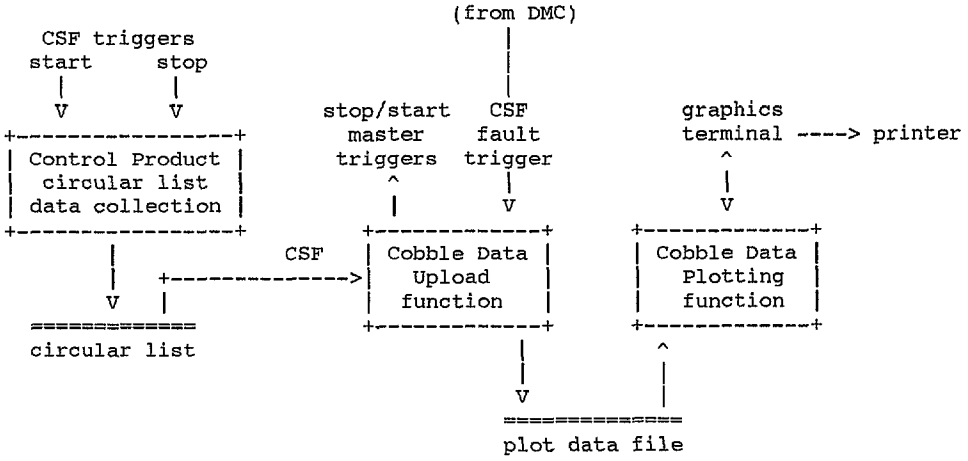
The "M" key aborts the current plot file interaction, and does NOT look for the next plot file, if any. Instead, this key aborts the entire function, returning the user to the menu (or DCL prompt) from which the program was entered.

3.3 Data Collection and Uploading

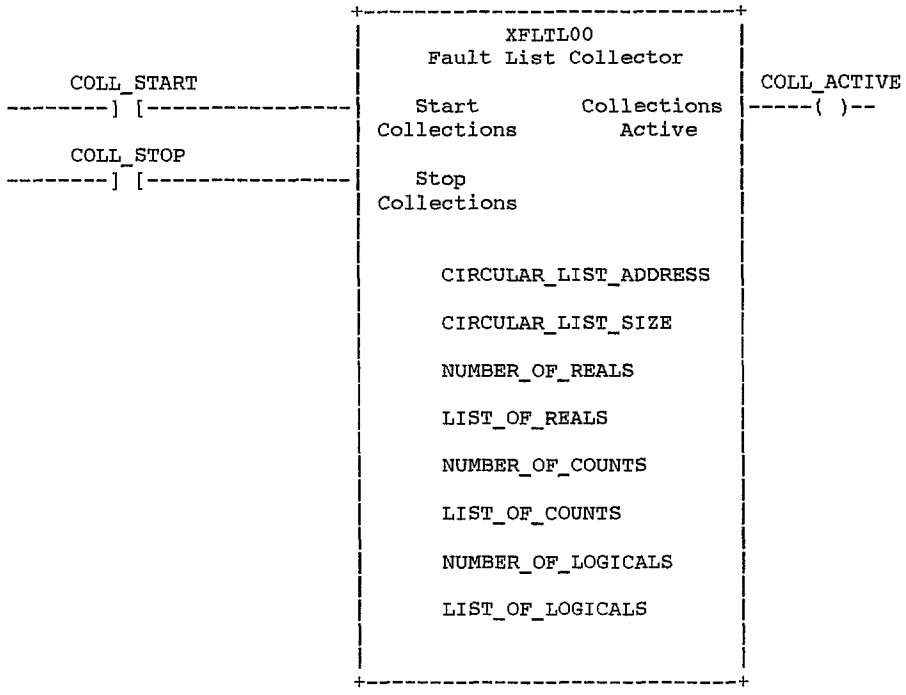
The data collection in each control product is independent of the control system, with the one exception that the control logic must provide a single "fault" output which will terminate the collection of data (the "fault trigger").

Each DMC requires a dedicated CBL segment which contains the data recording CBL block and any other special logic required to implement the data recording and fault triggering. The scan rate of this segment will determine the data collection rate. Two global CSF logicals (owned by the Supervisory VAX) are required to globally start and stop the data collection

in all control products. The following figure illustrates the data uploading function.



The cobble logging function is accomplished in DMC's by having a dedicated segment in each DMC with a single dedicated CBL block for fault data collection. The segment scan rate determines the frequency of fault data sampling. The block looks something like the following:



where LIST_OF_REALS is an array of DMC memory addresses where data is to be collected from, and NUMBER_OF_REALS is an integer counts between zero and 14 which specifies the number of addresses in the list or reals. NUMBER_OF_LOGICALS is the number of logical addresses in the LIST_OF_LOGICALS.

When the START_COLL input changes state from FALSE to TRUE, the block will initialize the circular list and begin sampling the locations specified in the three lists and storing the samples in its internal circular list. Data collection continues until the STOP_COLL bit transitions from FALSE to TRUE, at which time the block ceases collection of data and terminates the list. Only input transitions (from FALSE to TRUE) are processed by the block.

If the START_COLL and STOP_COLL inputs both transition from FALSE to TRUE on the same scan, the block will initialize the header but not take any samples.

4 Maintenance

4.1 Data Uploading Function

The upload function is primarily a background task which monitors the fault trigger(s) and uploads data when a fault occurs. It is driven by one or more change detect requests, and requires no user interface. The operator may also initiate an upload of the current data at any time.

Uploading from the DMC's is accomplished using directed messages (absolute memory requests) over the CSF. Each directed message will upload 238 bytes of data, so 69 directed messages will be required to upload the 16K circular list from each DMC.

The upload function is triggered by the CSF change detect function. A change detect request is issued for each stop trigger defined in the system. A stop trigger may be from a DMC or from any other CSF station. When one of the stop triggers is set, the upload function is activated by change detect. The upload function will set the CSF global STOP_COLL (after a small delay), which stops data collection in all DMC's and then begins uploading the circular list from each DMC. Uploaded data will be stored in files on the supervisory VAX disk.

The data upload function will interface with the data collection block by setting the "stop collection" CSF bit to suspend the collection of data. The data is then uploaded using CSF directed messages (which are handled by the DMC firmware and require no additional DMC code), and data collection is then automatically restarted by the upload function, which will set the "start collection" CSF bit.

4.2 Data Collection

The only interface between the DMC data collection software and other software is the two control bits (start and stop). These two triggers are both global CSF signals owned by the supervisory VAX, and used by all fault collection DMCs. An additional CSF bit is required by any CSF station which needs to provide a fault trigger to stop data collection.

4.2.1 Data Uploading Function

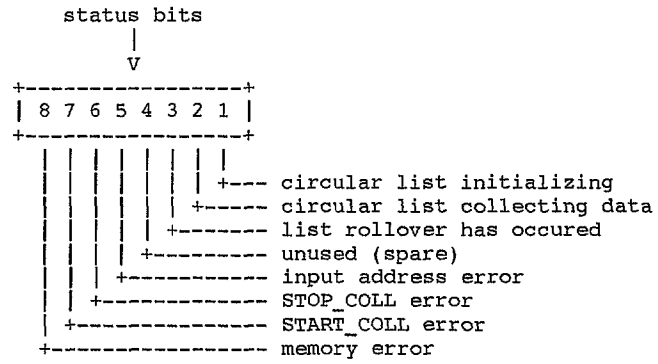
The VAX data collector maintains cobble data as VAX files which are not required to be "uploaded". This collector will sample data from IOCOM (any predefined set of CSF globals) and maintain an internal circular list of data in the same format as the DMC list. It uses the same CSF stop and start collection bits that the DMC upload function uses.

4.3 Error Handling

4.3.1 Data Collection

Errors during data collection, such as a bad address in the data address array, are handled by ignoring the invalid request(s) and processing all other valid requests. The 16K memory layout will reflect only the valid data which has been collected. If a section of the 16K memory map is invalid, the data from that control product will be invalidated and cannot be used in a plot.

The first two bytes of the circular list header will reflect the current state and error status of the data collection CBL block. The bits in the status byte are defined as follows:



The CBL block will set the "input address error" bit if one (or more) of the addresses in the input arrays to the block is (are) invalid, or if no data is defined to be collected (see below : r+c+g = 0). The following formula shows how to calculate the number of samples that will be collected for a 16K circular collection list.

$$\text{total_number_of_samples} = \frac{16384 - 16 - 12*(r+c+g)}{4*r + 2*c + g}$$

r = # of reals
c = # of counts
g = # of logicals

The "STOP_COLL error" bit is set if the CBL block is not collecting data and it gets a STOP_COLL request. Similarly, the "START_COLL error" bit is set if the block gets a START_COLL request when it is already collecting data.

Bit 1 will be 0 when the CBL block is not collecting data, and 1 when the block is initializing. Bit 2 will be set when the function is collecting data. The status word is cleared when the START_COLL input bit is asserted; this also sets bit 1 for a single pass (the initialization pass) and then clears bit 1 and sets bit 2 (collecting data) for all subsequent scans. Any errors which are detected will set the appropriate error bit.

4.3.2 Data Uploading Function

Errors during data upload will be sent to OPLOG. If the user is running the upload utility interactively, the errors will be displayed to the user. The upload program will restart the collection of data after all the circular lists have been uploaded. If the function cannot upload the data from one or more control products after several retry attempts, it will assume that the circular list data has been lost from that node and generate an appropriate message.

If any of the error flags in the uploaded status word are set, the user will be notified with an appropriate message.

5 Configuration

The Cobble Logging Utility does not require a major configuration to install. This utility is provided with defaults that should satisfy the majority of customers. Should the customer wish to alter the configuration for their site, the following items may be changed :

- Size of the circular list in each DMC ; normally 16K
- Number and type of variables in the VAX circular list ; 50 maximum
- DMC names from which data should be uploaded ; file driven.
- Number & Type (LOGICAL,COUNT,REAL) of variables sampled in each DMC.
- Layout and number of plot screens - see plot definition file description.

A Installation

The Cobble Logging function will be distributed on tape as a save set containing all necessary object modules to build the system. The customer will use the DEC VAX/VMS command procedure "VMSINSTAL" to install the cobble logging utility.